Data Preparation

Date	28 October 2026
Team ID	EN25CS3010016
Project Name	Global Olympic Sports Performance Analysis
Maximum Marks	5 Marks

Data preparation refers to the process of cleaning, standardizing, transforming, and organizing data so that it is suitable for analysis and reporting. In a Snowflake project, data preparation involves working with the ingested RAW data, identifying data quality issues, creating STAGING tables, applying required transformations, and preparing reliable data for the CURATED and MART layers.

Activity 1: Identifying and Handling Null Values

The four institutional CSV datasets were connected to Tableau for Global Olympic Sports Performance Analysis. The data was reviewed for missing values in relevant fields used for medal analysis, athlete participation, regional comparisons, and Olympic trends. Missing values should be assessed according to the affected field before deciding whether to retain, exclude, or handle them. No Snowflake queries were used because the project uses CSV files directly in Tableau.

Code / Procedure: Inspect the relevant fields in Tableau's Data Source view and review null values in the corresponding worksheets.

Output: Data fields were reviewed for suitability for visualization. Any missing values affecting the analysis should be documented and handled according to their meaning.

Activity 2: Duplicate Record Handling

The CSV datasets were reviewed to understand their structure and identify records that could potentially be repeated. Duplicate checking must account for the dataset's purpose: multiple medal records or appearances may be valid, so identical-looking rows should not automatically be deleted. No separate SQL-based duplicate-removal process was performed.

Code / Procedure: Review the dataset fields in Tableau and assess potential duplicates using appropriate record identifiers and the intended level of analysis.

Output: The datasets were connected for analysis. Any records identified as duplicates should be verified before removal to avoid affecting medal totals or athlete participation results.

Activity 3: Data Cleaning and Standardization

The four CSV datasets—`alltime_country_rankings.csv`, `athletes.csv`, `games_summary.csv`, and `medal_results.csv`—were connected to Tableau. Relevant fields and data types were reviewed for use in visualizations. Data relationships and field settings were configured to support country comparisons, medal trends, athlete gender analysis, sport participation, regional insights, and host-country comparisons.

Code / Procedure: Use Tableau's Data Source view to inspect field names, data types, and field values. Apply data-type or formatting corrections only where required.

Output: The datasets were prepared for visualization in Tableau, with relationships configured to support the project dashboards.

Activity 4: Data Transformation and Validation

Tableau was used to analyze the connected CSV datasets and build eight worksheets, three dashboards, and a four-point Story. Relationships between relevant datasets were configured, and interactive filters for Year, Season, and Sport were used. The calculated field `Unique Athletes` was created for athlete-related analysis. Dashboard filters and chart behavior were tested to confirm that the visualizations responded as intended.

Code / Procedure: Tableau calculated field used in the project:

```
Unique Athletes = COUNTD([Athlete ID])
```

Note: Use this expression only if your actual Tableau field is named `Athlete ID`; otherwise replace it with the exact athlete identifier field in your workbook.

Output: The dashboards and Story provide interactive views of Olympic medal performance, trends, athlete participation, regional comparisons, investment categories, and host-country performance.